

EDEXCEL INTERNATIONAL A LEVEL

# WMA12/01 May/June 2023 Worked Solutions

May/June 2023 paper rebuilt with the worked solution after each question.

**11**

paper questions

**WMA12**

Pure 2

**Single-  
paper  
solutions**standalone IAL  
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**PAST PAPER**

# **WMA12/01 May/June 2023**

**May/June 2023 | 11 questions | 75 marks**

**11**

questions

**75**

marks

**Answers**

worked solution  
after each  
question

## Question 1

## Integration

1. The continuous curve  $C$  has equation  $y = f(x)$ .

A table of values of  $x$  and  $y$  for  $y = f(x)$  is shown below.

|     |     |        |        |        |        |     |
|-----|-----|--------|--------|--------|--------|-----|
| $x$ | 4.0 | 4.2    | 4.4    | 4.6    | 4.8    | 5.0 |
| $y$ | 9.2 | 8.4556 | 3.8512 | 5.0342 | 7.8297 | 8.6 |

Use the trapezium rule with all the values of  $y$  in the table to find an approximation for

$$\int_4^5 f(x) \, dx$$

giving your answer to 3 decimal places.

(3)

# Worked Solution - Question 1

## 1. Identify the interval width

The  $x$  values go from 4.0 to 5.0 in steps of 0.2, so  $h = 0.2$ .

## 2. Apply the trapezium rule

$$\int_4^5 f(x) dx \approx \frac{0.2}{2} [9.2 + 8.6 + 2(8.4556 + 3.8512 + 5.0342 + 7.8297)].$$

## 3. Evaluate

The value is 6.81414, so the approximation is 6.814 to 3 decimal places.

**Final answer**

6.814.

## Question 2

## Polynomials

2.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

$$f(x) = 4x^3 - 8x^2 + 5x + a$$

where  $a$  is a constant.

Given that  $(2x - 3)$  is a factor of  $f(x)$ ,

(a) use the factor theorem to show that  $a = -3$  (2)

(b) Hence show that the equation  $f(x) = 0$  has only one real root. (4)

## Worked Solution - Question 2

**1. Use the factor theorem**

Since  $(2x - 3)$  is a factor,  $x = \frac{3}{2}$  is a root, so  $f\left(\frac{3}{2}\right) = 0$ .

**2. Substitute the root**

$$4\left(\frac{3}{2}\right)^3 - 8\left(\frac{3}{2}\right)^2 + 5\left(\frac{3}{2}\right) + a = 0.$$

**3. Find a**

$$13.5 - 18 + 7.5 + a = 0, \text{ so } 3 + a = 0 \text{ and } a = -3.$$

**4. Write the polynomial**

$$\text{With } a = -3, f(x) = 4x^3 - 8x^2 + 5x - 3.$$

**5. Factor using the known factor**

$$4x^3 - 8x^2 + 5x - 3 = (2x - 3)(2x^2 - x + 1).$$

**6. Check the quadratic**

$$\text{The discriminant of } 2x^2 - x + 1 \text{ is } (-1)^2 - 4(2)(1) = 1 - 8 = -7.$$

**7. Conclude about real roots**

Since the quadratic has negative discriminant, it has no real roots. Therefore

$$f(x) = 0 \text{ has only the one real root from } 2x - 3 = 0, \text{ namely } x = \frac{3}{2}.$$

**Final answer**

(a)  $a = -3$ .

(b)  $f(x) = 0$  has only one real root,  $x = \frac{3}{2}$ .

## Question 3

Circles

3. A circle  $C$  has centre  $(2, 5)$

Given that the point  $P(8, -3)$  lies on  $C$

(a) (i) find the radius of  $C$

(ii) find an equation for  $C$

(3)

(b) Find the equation of the tangent to  $C$  at  $P$  giving your answer in the form  $ax + by + c = 0$  where  $a$ ,  $b$  and  $c$  are integers to be found.

(4)



## Worked Solution - Question 3

**1. Use the distance formula**

The centre is  $(2, 5)$  and  $P = (8, -3)$ . The radius is  $\sqrt{(8-2)^2 + (-3-5)^2}$ .

**2. Find the radius**

$$r = \sqrt{6^2 + (-8)^2} = \sqrt{100} = 10.$$

**3. Write the circle equation**

With centre  $(2, 5)$  and radius 10, the circle is  $(x-2)^2 + (y-5)^2 = 100$ .

**4. Find the radius gradient**

The gradient of the radius from the centre to  $P$  is  $\frac{-3-5}{8-2} = \frac{-8}{6} = -\frac{4}{3}$ .

**5. Find the tangent gradient**

The tangent is perpendicular to the radius, so its gradient is  $\frac{3}{4}$ .

**6. Use point P**

The tangent through  $(8, -3)$  is  $y + 3 = \frac{3}{4}(x - 8)$ .

**7. Put into integer form**

Multiplying by 4 gives  $4y + 12 = 3x - 24$ , hence  $3x - 4y - 36 = 0$ .

**Final answer**

*(a)(i)* 10.

*(a)(ii)*  $(x - 2)^2 + (y - 5)^2 = 100$ .

*(b)*  $3x - 4y - 36 = 0$ .

## Question 4

## Binomial Expansion

4. The binomial expansion, in ascending powers of  $x$ , of

$$(3 + px)^5$$

where  $p$  is a constant, can be written in the form

$$A + Bx + Cx^2 + Dx^3 \dots$$

where  $A$ ,  $B$ ,  $C$  and  $D$  are constants.

- (a) Find the value of  $A$

(1)

Given that

- $B = 18D$
- $p < 0$

- (b) find

- (i) the value of  $p$
- (ii) the value of  $C$

(6)

## Worked Solution - Question 4

**1. Find the constant term**

In  $(3 + px)^5$ , the constant term is  $3^5 = 243$ , so  $A = 243$ .

**2. Find B**

The coefficient of  $x$  is  $5(3^4)p = 405p$ , so  $B = 405p$ .

**3. Find D**

The coefficient of  $x^3$  is  $\binom{5}{3}(3^2)p^3 = 10 \cdot 9p^3 = 90p^3$ , so  $D = 90p^3$ .

**4. Use B equals 18D**

$$405p = 18(90p^3) = 1620p^3.$$

**5. Solve for p**

Since  $p < 0$ ,  $p \neq 0$ . Divide by  $405p$  to get  $1 = 4p^2$ , so  $p = \pm \frac{1}{2}$ . Therefore  $p = -\frac{1}{2}$ .

**6. Find C**

The coefficient of  $x^2$  is  $\binom{5}{2}(3^3)p^2 = 10 \cdot 27 \cdot \frac{1}{4} = \frac{135}{2}$ .

**Final answer**

(a)  $A = 243$ .

(b)(i)  $p = -\frac{1}{2}$ .

(b)(ii)  $C = \frac{135}{2}$ .

## Question 5

### Laws of Logarithms

5. Use the laws of logarithms to solve

$$\log_2(16x) + \log_2(x + 1) = 3 + \log_2(x + 6)$$

(5)

## Worked Solution - Question 5

Topic group

**1. State the domain**

The logarithms require  $16x > 0$ ,  $x + 1 > 0$  and  $x + 6 > 0$ , so  $x > 0$ .

**2. Combine the left side**

$$\log_2(16x) + \log_2(x + 1) = \log_2(16x(x + 1)).$$

**3. Rewrite the right side**

$$3 = \log_2 8, \text{ so } 3 + \log_2(x + 6) = \log_2(8(x + 6)).$$

**4. Equate the arguments**

$$\text{Therefore } 16x(x + 1) = 8(x + 6).$$

**5. Solve the quadratic**

$$16x^2 + 16x = 8x + 48, \text{ so } 16x^2 + 8x - 48 = 0. \text{ Dividing by 8 gives } 2x^2 + x - 6 = 0.$$

**6. Factor and use the domain**

$$(2x - 3)(x + 2) = 0, \text{ so } x = \frac{3}{2} \text{ or } x = -2. \text{ The domain } x > 0 \text{ rejects } -2, \text{ leaving } x = \frac{3}{2}.$$

**Final answer**

$$x = \frac{3}{2}.$$

## Question 6

## Geometric Sequences

6. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

A software developer released an app to download.

The numbers of downloads of the app each month, in thousands, for the first three months after the app was released were

$$2k - 15 \quad k \quad k + 4$$

where  $k$  is a constant.

Given that the numbers of downloads each month are modelled as a geometric series,

(a) show that  $k^2 - 7k - 60 = 0$  (2)

(b) predict the number of downloads in the 4th month. (4)

The **total** number of all downloads of the app is predicted to exceed 3 million for the first time in the  $N$ th month.

(c) Calculate the value of  $N$  according to the model. (3)

# Worked Solution - Question 6

## 1. Use the geometric condition

For three consecutive terms of a geometric sequence, the middle term squared equals the product of the outer terms.

## 2. Set up the equation

So  $k^2 = (2k - 15)(k + 4)$ .

## 3. Show the required quadratic

Expanding gives  $k^2 = 2k^2 - 7k - 60$ , hence  $k^2 - 7k - 60 = 0$ .

## 4. Find $k$

$k^2 - 7k - 60 = (k - 12)(k + 5) = 0$ , so  $k = 12$  or  $k = -5$ . The download numbers must be positive, so  $k = 12$ .

## 5. Find the common ratio

The first three monthly values are **9**, **12** and **16** thousand. The common ratio is  $\frac{12}{9} = \frac{4}{3}$ .

## 6. Predict month four

The 4th month is  $16 \cdot \frac{4}{3} = \frac{64}{3}$  thousand downloads, which is about **21300** downloads.

## 7. Write the cumulative total

In thousands, the total after  $N$  months is

$$S_N = \frac{9 \left( \left( \frac{4}{3} \right)^N - 1 \right)}{\frac{4}{3} - 1} = 27 \left( \left( \frac{4}{3} \right)^N - 1 \right).$$



### 8. Compare with three million

Three million downloads is **3000** thousand, so we need

$$27 \left( \left( \frac{4}{3} \right)^N - 1 \right) > 3000.$$

### 9. Solve the inequality

$$\left( \frac{4}{3} \right)^N > \frac{1009}{9}. \text{ Taking logarithms gives } N > \frac{\log(1009/9)}{\log(4/3)} = 16.405 \dots$$

### 10. Choose the first month

The first integer month for which the total exceeds **3** million is  $N = 17$ .

#### Final answer

(a) Proven.

(b)  $\frac{64}{3}$  thousand downloads, about **21300** downloads.

(c)  $N = 17$ .

## Question 7

## Applications of Differentiation

7. The height of a river above a fixed point on the riverbed was monitored over a 7-day period.

The height of the river,  $H$  metres,  $t$  days after monitoring began, was given by

$$H = \frac{\sqrt{t}}{20}(20 + 6t - t^2) + 17 \quad 0 \leq t \leq 7$$

Given that  $H$  has a stationary value at  $t = \alpha$

- (a) use calculus to show that  $\alpha$  satisfies the equation

$$5\alpha^2 - 18\alpha - 20 = 0 \quad (5)$$

- (b) Hence find the value of  $\alpha$ , giving your answer to 3 decimal places. (1)

- (c) Use further calculus to prove that  $H$  is a maximum at this value of  $\alpha$ . (2)

# Worked Solution - Question 7

Topic group

**1. Rewrite H with powers**

$$H = \frac{1}{20}t^{1/2}(20 + 6t - t^2) + 17.$$

**2. Expand before differentiating**

$$H = \frac{1}{20}(20t^{1/2} + 6t^{3/2} - t^{5/2}) + 17.$$

**3. Differentiate**

$$\frac{dH}{dt} = \frac{1}{20} \left( 10t^{-1/2} + 9t^{1/2} - \frac{5}{2}t^{3/2} \right).$$

**4. Put over a common factor**

$$\frac{dH}{dt} = \frac{-5t^2 + 18t + 20}{40\sqrt{t}}.$$

**5. Use the stationary condition**

At  $t = \alpha$ ,  $\frac{dH}{dt} = 0$ . Since  $\alpha > 0$ , the denominator is not zero, so  
 $-5\alpha^2 + 18\alpha + 20 = 0$ .

**6. Show the required equation**

Multiplying by  $-1$  gives  $5\alpha^2 - 18\alpha - 20 = 0$ .

**7. Find alpha**

$$\alpha = \frac{18 \pm \sqrt{18^2 + 4(5)(20)}}{10} = \frac{9 \pm \sqrt{181}}{5}.$$

### 8. Use the time interval

Only the positive value lies in  $0 \leq t \leq 7$ , so  $\alpha = \frac{9 + \sqrt{181}}{5} = 4.491$  to 3 decimal places.

### 9. Differentiate again

$$\frac{d^2 H}{dt^2} = \frac{-15t^2 + 18t - 20}{80t^{3/2}}.$$

### 10. Use the stationary equation

At  $t = \alpha$ ,  $5\alpha^2 = 18\alpha + 20$ . Therefore the numerator of  $\frac{d^2 H}{dt^2}$  is  $-15\alpha^2 + 18\alpha - 20 = -3(18\alpha + 20) + 18\alpha - 20 = -36\alpha - 80$ .

### 11. Show it is negative

Since  $\alpha > 0$ ,  $-36\alpha - 80 < 0$ , and the denominator  $80\alpha^{3/2}$  is positive. Thus  $\frac{d^2 H}{dt^2} < 0$ , so  $H$  is a maximum.

#### Final answer

(a) Proven.

(b)  $\alpha = 4.491$ .

(c)  $H$  is a maximum at this value.

## Question 8

Proof

8. (i) A student writes the following statement:

“When  $a$  and  $b$  are consecutive **prime** numbers,  $a^2 + b^2$  is never a multiple of 10”

Prove by counter example that this statement is **not** true.

(2)

- (ii) Given that  $x$  and  $y$  are even integers greater than 0 and less than 6, prove by exhaustion, that

$$1 < x^2 - \frac{xy}{4} < 15$$

(3)

# Worked Solution - Question 8

Topic group

**1. Choose consecutive primes**

7 and 11 are consecutive prime numbers.

**2. Check the statement**

$7^2 + 11^2 = 49 + 121 = 170$ , which is a multiple of 10.

**3. Conclude the counterexample**

This single counterexample proves the student's statement is not true.

**4. List the possible even values**

The even integers greater than 0 and less than 6 are 2 and 4. Therefore the possible ordered pairs are (2, 2), (2, 4), (4, 2), (4, 4).

**5. Check x equals two**

If  $(x, y) = (2, 2)$ , then  $x^2 - \frac{xy}{4} = 4 - 1 = 3$ . If  $(x, y) = (2, 4)$ , then  $x^2 - \frac{xy}{4} = 4 - 2 = 2$ .

**6. Check x equals four**

If  $(x, y) = (4, 2)$ , then  $x^2 - \frac{xy}{4} = 16 - 2 = 14$ . If  $(x, y) = (4, 4)$ , then  $x^2 - \frac{xy}{4} = 16 - 4 = 12$ .

**7. Complete the exhaustion**

The possible values are 3, 2, 14, 12, and each lies strictly between 1 and 15. Hence  $1 < x^2 - \frac{xy}{4} < 15$  in every possible case.

**Final answer**

(i) A counterexample is  $a = 7, b = 11$ , since  $7^2 + 11^2 = 170$ .

(ii) Exhaustion proves the inequality for  
 $(x, y) = (2, 2), (2, 4), (4, 2), (4, 4)$ .

## Question 9

## Trigonometric Equations

9. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(a) Show that

$$3 \cos \theta (\tan \theta \sin \theta + 3) = 11 - 5 \cos \theta$$

may be written as

$$3 \cos^2 \theta - 14 \cos \theta + 8 = 0 \quad (3)$$

(b) Hence solve, for  $0 < x < 360^\circ$

$$3 \cos 2x (\tan 2x \sin 2x + 3) = 11 - 5 \cos 2x$$

giving your answers to one decimal place. (4)



# Worked Solution - Question 9

Topic group

**1. Rewrite  $\tan \theta \sin \theta$** 

$$\tan \theta \sin \theta = \frac{\sin^2 \theta}{\cos \theta}.$$

**2. Simplify the left side**

$$3 \cos \theta (\tan \theta \sin \theta + 3) = 3 \sin^2 \theta + 9 \cos \theta.$$

**3. Use sine squared**

$$\text{The equation becomes } 3(1 - \cos^2 \theta) + 9 \cos \theta = 11 - 5 \cos \theta.$$

**4. Rearrange**

$$3 - 3 \cos^2 \theta + 9 \cos \theta = 11 - 5 \cos \theta, \text{ so } 3 \cos^2 \theta - 14 \cos \theta + 8 = 0.$$

**5. Apply the result to  $2x$** 

$$\text{For the equation in part (b), take } \theta = 2x. \text{ Then } 3 \cos^2(2x) - 14 \cos(2x) + 8 = 0.$$

**6. Solve the quadratic**

$$\text{Let } u = \cos(2x). \text{ Then } 3u^2 - 14u + 8 = 0, \text{ so } (3u - 2)(u - 4) = 0.$$

**7. Reject the impossible value**

$$u = 4 \text{ is impossible for cosine, so } \cos(2x) = \frac{2}{3}.$$

**8. Use the interval**

$$\text{Since } 0 < x < 360^\circ, \text{ we have } 0 < 2x < 720^\circ.$$

**9. Find  $x$** 

$$\text{In this interval, } 2x = 48.189 \dots^\circ, 311.810 \dots^\circ, 408.189 \dots^\circ, 671.810 \dots^\circ.$$

$$\text{Dividing by 2 gives } x = 24.1^\circ, 155.9^\circ, 204.1^\circ, 335.9^\circ \text{ to one decimal place.}$$

**Final answer**

(a) Proven.

(b)  $x = 24.1^\circ, 155.9^\circ, 204.1^\circ, 335.9^\circ$ .

## Question 10

## Integration

10. The curve  $C$  has equation

$$y = \frac{(x - k)^2}{\sqrt{x}} \quad x > 0$$

where  $k$  is a **positive** constant.

(a) Show that

$$\int_1^{16} \frac{(x - k)^2}{\sqrt{x}} \, dx = ak^2 + bk + \frac{2046}{5}$$

where  $a$  and  $b$  are integers to be found.

(5)

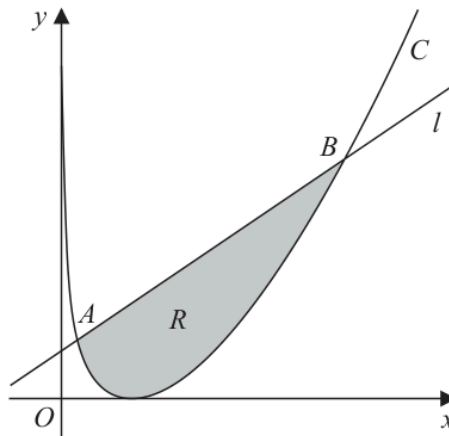


Figure 1

Figure 1 shows a sketch of the curve  $C$  and the line  $l$ .

Given that  $l$  intersects  $C$  at the point  $A(1, 9)$  and at the point  $B(16, q)$  where  $q$  is a constant,

(b) show that  $k = 4$

(2)

The region  $R$ , shown shaded in Figure 1, is bounded by  $C$  and  $l$

Using the answers to parts (a) and (b),

(c) find the area of region  $R$

(3)

## Worked Solution - Question 10

**1. Expand the integrand**

$$\frac{(x-k)^2}{\sqrt{x}} = \frac{x^2 - 2kx + k^2}{x^{1/2}} = x^{3/2} - 2kx^{1/2} + k^2x^{-1/2}.$$

**2. Integrate**

$$\int (x^{3/2} - 2kx^{1/2} + k^2x^{-1/2}) dx = \frac{2}{5}x^{5/2} - \frac{4k}{3}x^{3/2} + 2k^2x^{1/2}.$$

**3. Substitute the limits**

From 1 to 16, this gives  $\left(\frac{2048}{5} - \frac{256k}{3} + 8k^2\right) - \left(\frac{2}{5} - \frac{4k}{3} + 2k^2\right).$

**4. Simplify the integral**

The result is  $6k^2 - 84k + \frac{2046}{5}$ , so  $a = 6$  and  $b = -84$ .

**5. Use point A**

Since  $A(1, 9)$  lies on  $C$ ,  $9 = \frac{(1-k)^2}{\sqrt{1}} = (1-k)^2.$

**6. Find k**

Thus  $1 - k = \pm 3$ , so  $k = 4$  or  $k = -2$ . Since  $k$  is positive,  $k = 4$ .

**7. Find point B**

When  $k = 4$  and  $x = 16$ ,  $q = \frac{(16-4)^2}{\sqrt{16}} = \frac{144}{4} = 36$ , so  $B = (16, 36).$

**8. Find the line**

The line through  $A(1, 9)$  and  $B(16, 36)$  has gradient  $\frac{36-9}{16-1} = \frac{9}{5}$ , so  $l$  has equation  $y = \frac{9}{5}x + \frac{36}{5}.$

### 9. Area under the line

$$\int_1^{16} \left( \frac{9}{5}x + \frac{36}{5} \right) dx = \left[ \frac{9}{10}x^2 + \frac{36}{5}x \right]_1^{16} = \frac{675}{2}.$$

### 10. Area under the curve

From part (a), with  $k = 4$ ,  $\int_1^{16} \frac{(x-4)^2}{\sqrt{x}} dx = 6(4)^2 - 84(4) + \frac{2046}{5} = \frac{846}{5}.$

### 11. Subtract curve from line

The shaded area is  $\frac{675}{2} - \frac{846}{5} = \frac{1683}{10}.$

#### Final answer

(a)  $a = 6$ ,  $b = -84$ .

(b)  $k = 4$ .

(c) Area =  $\frac{1683}{10}.$

## Question 11

## Sequences &amp; Series

11. A sequence  $u_1, u_2, u_3, \dots$  is defined by

$$u_{n+1} = b - au_n$$

$$u_1 = 3$$

where  $a$  and  $b$  are constants.

(a) Find, in terms of  $a$  and  $b$ ,

(i)  $u_2$

(ii)  $u_3$

(2)

Given

- $\sum_{n=1}^3 u_n = 153$

- $b = a + 9$

(b) show that

$$a^2 - 5a - 66 = 0$$

(3)

(c) Hence find the larger possible value of  $u_2$

(3)

# Worked Solution - Question 11

**1. Find  $u_2$** 

Using  $u_{n+1} = b - au_n$  and  $u_1 = 3$ , we get  $u_2 = b - a(3) = b - 3a$ .

**2. Find  $u_3$** 

$$u_3 = b - au_2 = b - a(b - 3a) = b - ab + 3a^2.$$

**3. Use the sum condition**

$$u_1 + u_2 + u_3 = 153, \text{ so } 3 + (b - 3a) + (b - ab + 3a^2) = 153.$$

**4. Substitute  $b$  equals  $a$  plus nine**

Using  $b = a + 9$ , the left side becomes

$$3 + (a + 9) - 3a + (a + 9) - a(a + 9) + 3a^2.$$

**5. Simplify**

$$\text{This is } 2a^2 - 10a + 21, \text{ so } 2a^2 - 10a + 21 = 153.$$

**6. Show the required equation**

$$2a^2 - 10a - 132 = 0. \text{ Dividing by 2 gives } a^2 - 5a - 66 = 0.$$

**7. Solve for  $a$** 

$$a^2 - 5a - 66 = (a - 11)(a + 6) = 0, \text{ so } a = 11 \text{ or } a = -6.$$

**8. Find the corresponding  $u_2$  values**

If  $a = 11$ , then  $b = 20$  and  $u_2 = 20 - 3(11) = -13$ . If  $a = -6$ , then  $b = 3$  and  $u_2 = 3 - 3(-6) = 21$ .

**9. Choose the larger value**

The larger possible value of  $u_2$  is 21.

**Final answer**

(a)(i)  $u_2 = b - 3a$ .

(a)(ii)  $u_3 = b - ab + 3a^2$ .

(b) Proven.

(c) 21.